Bipedal Locomotion Control based on Divergent Component of Motion

First Mini-Project of Legged Robots

Introduction

In this project you will work on optimization-based control of bipedal robots by using the Divergent Component of Motion (DCM) concept. The code structure will include three parts: the DCM planning and optimization, Foot Trajectory Planner, and Inverse Kinematics. The only block that you need to implement is the DCM planning and optimization. In other words, you only need to open and edit the `DCMTrajectoryGenerator.py` class and implement the quadratic programming optimization in the Jupyter Notebook and follow the comments that have been written in the codes (it is specified by #todo). You can find the code by the following link:

https://gitlab.epfl.ch/shafiee/lr-dcm-start.git

1 Method

1.1 Linear Inverted Pendulum Model (LIPM)

The LIPM has been widely utilized to describe the dynamics of the CoM for bipedal locomotion [3]. The LIPM assumes a constant rate of change of centroidal angular momentum and movement of the CoM height within a plane. These assumptions are suitable for planning humanoid robot locomotion, as research on human walking indicates minimal variations in centroidal angular momentum and CoM height [1]. Based on the assumptions, the equations of motion for the LIPM can be derived as follows:

$$\ddot{\mathbf{x}} = \omega^2(\mathbf{x} - \mathbf{cop}) \tag{1}$$

in which $\mathbf{x} = [x_{com}, y_{com}]^T$ is the horizontal position of the CoM, $\omega_0 = \sqrt{\frac{g}{\Delta z}}$ is the natural frequency of the LIPM, and $\mathbf{cop} = [cop_x, cop_y]^T$ is the horizontal position of the center of pressure (CoP).

1.2 Divergent Component of Motion

In this section, we provide an overview of the DCM concept's background. The dynamics of the CoM, as modeled by the LIPM, can be split into stable and unstable components. The unstable component is referred to as the DCM and is defined as follows:

$$\boldsymbol{\xi} = \mathbf{x} + \frac{\dot{\mathbf{x}}}{\omega} \tag{2}$$

Throughout this study, DCM is represented as ξ as the notation in [5]. From (2), the CoM dynamics is given by:

$$\dot{\mathbf{x}} = \omega(\boldsymbol{\xi} - \mathbf{x}) \tag{3}$$

By differentiating (2) and substituting (1), the DCM dynamics is expressed as :

$$\dot{\boldsymbol{\xi}} = \omega(\boldsymbol{\xi} - \mathbf{cop}) \tag{4}$$

Fig. (1) illustrates the relationship between DCM dynamics, CoM, and the CoP. By re-arranging DCM dynamics (4), the following ordinary differential equation (ODE) holds:

$$\dot{\boldsymbol{\xi}} - \omega\boldsymbol{\xi} = -\omega\,\mathbf{cop}_0 \tag{5}$$

The solution to (5) writes:

$$\boldsymbol{\xi}(t) = e^{\int \omega dt}\left[\int (-\mathbf{cop}_0\,\omega)\, e^{\int -\omega dt} dt + \mathbf{C}\right], \tag{6}$$

where $C \in \mathbb{R}^2$ is the vector of unknown coefficients that can be found by imposing the boundary conditions. Therefore, we can find these coefficients by solving the problem (6) either as an initial value problem, namely

$$\boldsymbol{\xi}(0) = \boldsymbol{\xi}_0 = \mathbf{cop}_0 + \mathbf{C}_0, \tag{7}$$

or as a final value problem:

$$\boldsymbol{\xi}(T) = \boldsymbol{\xi}_T = \mathbf{cop_0} + \mathbf{C_f}\, e^{\omega T}. \tag{8}$$

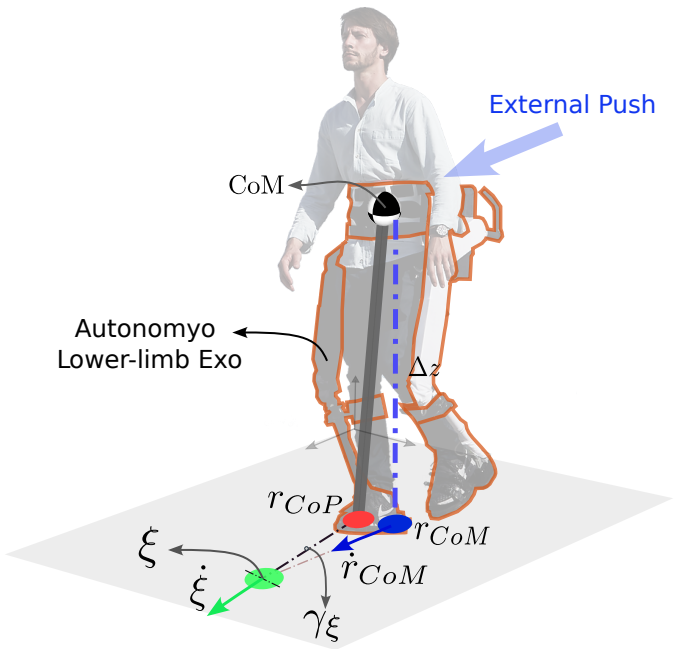

Figure 1: DCM, CoM and CoP Points correlations for Centroidal Dynamics

Therefore, by solving the equation (4) as an initial value problem, we arrive at the following equation that represents the time evolution of the DCM:

$$\boldsymbol{\xi} = (\boldsymbol{\xi}_0 - \mathbf{cop_0})\exp(\omega t) + \mathbf{cop_0} \tag{9}$$

We also can solve the CoM dynamics (3) by treating it as an initial value problem:

$$\mathbf{x} = (\mathbf{x_0} - \boldsymbol{\xi}_0)\exp(-\omega t) + \boldsymbol{\xi}_0 \tag{10}$$

As evident from the above equation, the CoM exhibits stable dynamics, with the exponential term being negative. However, the DCM exhibits unstable dynamics, characterized by a positive exponential term. This indicates that the difference between ξ_0 and cop_0 increases exponentially over time. Therefore, a prerequisite for ensuring the stability of the CoM trajectory is that the DCM trajectory remains stable. As stated in [2], the concept of DCM provides a relatively straightforward approach for formulating stability requirements in walking. A simple rule proves effective in ensuring walking stability: when placing the foot, position the Center of Pressure (CoP) at a certain distance behind and outward from the DCM at the moment of foot contact. This distance between the CoP and the DCM during foot placement is referred to as the DCM offset, and minimizing this distance is crucial for maintaining viable states.

1.3 Push Recovery Controller

This section presents the step adaption mechanism based on DCM dynamics [4]. More precisely, we present below a step adjustment strategy that optimizes the next step position and timing based on the measured DCM.

To find a DCM trajectory that satisfies both the initial and the final condition problems, the coefficient C_0 must equal C_f. Thus, by combining (7) and (8), one has:

$$\boldsymbol{\xi}_0 - \mathbf{cop}_0 = (\boldsymbol{\xi}_T - \mathbf{cop}_0)\, e^{-\omega T}. \tag{11}$$

Now by defining $\sigma = e^{\omega T}$ we obtain :

$$\boldsymbol{\xi}_T + \mathbf{cop}_0(-1+\sigma) - \boldsymbol{\xi}_0\sigma = 0. \tag{12}$$

Let $\mathbf{cop}_T$ represent the CoP position at the start of the next step, and $\boldsymbol{\gamma_T} = \boldsymbol{\xi_T} - \mathbf{cop_T}$ denote the DCM offset for the next step (i.e, the end of this step). Therefore, straightforward calculations lead to:

$$\boldsymbol{\gamma_T} + \mathbf{cop}_T + (\mathbf{cop}_0 - \boldsymbol{\xi}_0)\,\sigma = \mathbf{cop}_0. \tag{13}$$

The step adjustment problem can be formalized as a constrained optimization problem, wherein the search variables consist of γ_T, $\mathbf{cop}_T$, and σ, and the cost function is appropriately defined in a quadratic manner. It's worth noting that the desired final DCM position and step timing are dependent on γ_T and σ, respectively. Additionally, $\mathbf{cop}_T$ is assumed to be located at the center of the foot at the start of the next step. Therefore, we can treat this position as the target for the upcoming footstep placement. The selected cost function aims to minimize the deviation of the desired gait values from the nominal ones:

$$
\begin{aligned}
J = \alpha_1 |cop_{xT} - cop_{xT,nom}|^2 + \alpha_2 |cop_{yT} - cop_{yT,nom}|^2 + \alpha_3 |\gamma_{T,x} - \gamma_{nom,x}|^2 + \alpha_4 |\gamma_{T,y} - \gamma_{nom,y}|^2 \\
+ \alpha_5 |\sigma - e^{\omega T_{nom}}|^2,
\end{aligned} \tag{14}
$$

where α_1, α_2, α_3 are positive numbers and the next ZMP position $\mathbf{cop}_{T,nom}$, step duration T_{nom} and next DCM offset γ_{nom} are the desired values.

We also present the following set of inequality constraints:

$$
\begin{bmatrix} I_2 & 0_{2\times1} & 0_2 \\ -I_2 & 0_{2\times1} & 0_2 \\ 0_{1\times2} & I_1 & 0_{1\times2} \\ 0_{1\times2} & -I_1 & 0_{1\times2} \end{bmatrix} \begin{bmatrix} \mathbf{cop}_T \\ \sigma \\ \gamma_T \end{bmatrix} \leq \begin{bmatrix} \mathbf{cop}_{T,max} \\ -\mathbf{cop}_{T,min} \\ \sigma_{max} \\ -\sigma_{min} \end{bmatrix}, \tag{15}
$$

Here, $\mathbf{cop}_{T,max}$ and $\mathbf{cop}_{T,min}$ are in $\mathbb{R}^2$, while σ_{max} and σ_{min} belong to $\mathbb{R}$. These inequality constraints are established considering the constraints imposed by leg kinematics on the maximum step length and by the maximum achievable velocity on the minimum step duration. Lastly, the relationship described in (13) is considered as an equality constraint. It is worth noting that during push recovery, the controller attempts to minimize the DCM offset to have a zero dcm offset. Due to the quadratic and linear dependence of the cost function and constraints on the unknown variables, the entire framework can be formulated as a Quadratic Programming (QP) problem:

$$
\begin{aligned}
\underset{x}{\text{minimize}} \quad & \tfrac{1}{2} x^T P x + q^T x \\
\text{subject to} \quad & Gx \leq h \\
& Ax = b
\end{aligned} \tag{16}
$$

Thus, the equations 14, 15, 13 should be written in the form of equation 16 and QP library can be used to solve this problem.

Report

Please upload your code, a video of your results, and create a short report (less than 5 pages) including details on the method, plots, and answers to the following questions:

Questions

1. Based on equation (9), which physical parameters will affect the rate of divergence of the DCM dynamics?
2. In the optimization-based DCM motion planning, how do we consider the stability of DCM movement?
3. You will observe that the controller will not be able to recover the balance of the robot by changing the nominal given parameters for optimization and feet trajectory parameters. What are the reason for this observation? which assumption during the controller implementation leads to this problem?
4. What other balance recovery strategies can be used in addition to the stepping strategy? (Think about your balance recovery strategies, how do you recover from external disturbances in different situations?)
5. What's the reason that we use the DCM equations in the optimization for balance recovery? Why we do not use the whole dynamic equation of the robot to solve the control problem?
6. (**Bonus**) Implement a push recovery strategy **during locomotion**, and optimize the step position and duration based on external push for atlas humanoid robot. (You could get inspiration from the first part of the project that was about CoM motion planning.)

References

[1] Hugh Herr and Marko Popovic. Angular momentum in human walking. *Journal of experimental biology*, 211(4):467–481, 2008.

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[3] Shuuji Kajita, Fumio Kanehiro, Kenji Kaneko, Kiyoshi Fujiwara, Kensuke Harada, Kazuhito Yokoi, and Hirohisa Hirukawa. Biped walking pattern generation by using preview control of zero-moment point. In *International Conference on Robotics and Automation*, volume 2, pages 1620–1626. IEEE, 2003.

[4] Milad Shafiee, Giulio Romualdi, Stefano Dafarra, Francisco Javier Andrade Chavez, and Daniele Pucci. Online dcm trajectory generation for push recovery of torque-controlled humanoid robots. In *International Conference on Humanoid Robots (Humanoids)*, pages 671–678. IEEE, 2019.

[5] Toru Takenaka, Takashi Matsumoto, and Takahide Yoshiike. Real time motion generation and control for biped robot-1 st report: Walking gait pattern generation. In *International Conference on Intelligent Robots and Systems*, pages 1084–1091. IEEE, 2009.